

CSE 114

Introduction to Object-Oriented Programming

Course Information

CSE 114
Introduction to Object-Oriented Programming
Stony Brook University
Instructor: Ahmad Esmaili
<http://www.cs.stonybrook.edu/~cse114>

Instructor Information

- Ahmad Esmaili
210 New Computer Science Building
- Office hours: Mondays & Wednesdays 8:20-9:20AM or
By Appointment
- Email: esmaili@cs.stonybrook.edu
- TAs: see course Web page:
<http://www.cs.stonybrook.edu/~cse114>
- Please include “CSE 114” in the email subject and your name in
your email correspondence

Course Description

- “An introduction to procedural and object-oriented programming methodology. Topics include program structure, conditional and iterative programming, procedures, arrays and records, object classes, encapsulation, information hiding, inheritance, polymorphism, file I/O, and exceptions. Includes required laboratory. This course has been designated as a High Demand/Controlled Access (HD/CA) course. Students registering for HD/CA courses for the first time will have priority to do so.”

General Information

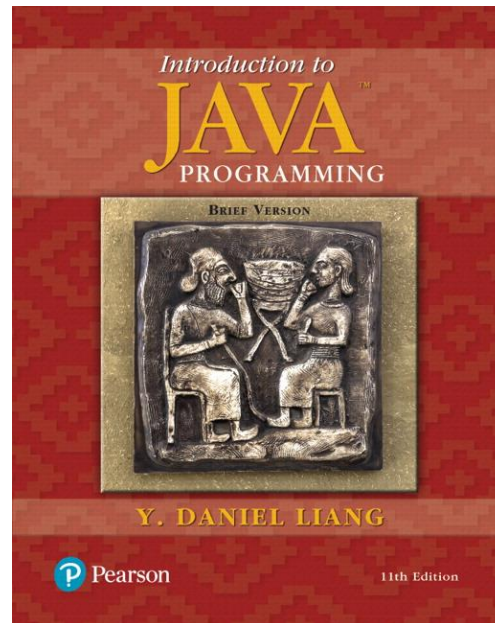
- Meeting Information:
- Lectures: Monday, Wednesday 9:30am – 12:45pm online Synchronous.
- Labs: Monday, Wednesday 1 – 4pm online Synchronous.
- Course Web page: <http://www.cs.stonybrook.edu/~cse114>
- **Brightspace** will be used for discussion board, assignments, lab material, lab grades and course material

Correction Day:

- Friday, May 29 - Classes follow a Monday schedule.
No classes on May 25 (Memorial Day).

Textbook

- Introduction To Java Programming, Brief Version,
Author: Liang, Publisher: Pearson, Edition: 11th



- Necessary Software:
 - Java 1.21: download from
<http://www.oracle.com/technetwork/java/javase/downloads>
 - Eclipse IDE: <http://www.eclipse.org>

Course Focus

- Introduction to programming (in Java): conditional statements, loops
- Introduce the basic concepts of object-oriented programming
 - object classes
 - encapsulation
 - inheritance
 - polymorphism
 - Application: GUIs
- Fundamental data structures of high-level programming: arrays, lists, stacks, ...
- Algorithms
- Programming assignments
 - systematic software debugging techniques
 - systematic testing techniques

Major Course topics

1. Procedural Programming Basics:

- variable declarations
- data types
- assignment statements & expressions
- textual manipulation & strings
- input/output
- method construction
- conditional (branching) statements
- iteration = loops and methods

Major Course topics

2. Arrays:

- collect data in arrays
- searching
- sorting
- array manipulations

3. Object Oriented Programming:

- designing and constructing classes using containment
- aggregation
- inheritance
- polymorphism
- *Application: GUIs*

Coursework

● Grading Schema

● Final Exam:	37%
● Midterm Exam	33%
● 5 Programming Assignments:	14%
● Lab Attendance and Exercises:	10%
● Pop quiz:	<u>6%</u>
	100%
● Revel (MyLab MyProgrammigLab)	?? %

Special Rule: If all grades are above the respective class averages, you're guaranteed to receive a C or higher in this class.

Exams Dates

- Midterm: Wednesday, June 17th, 9:30am-12:00pm
- Final Exam: Wednesday July 1st, 1:30-4:00pm

No make-up exams!

Online Exams

- Using Respondus LockDown Browser, webcam & mic
- **Single attempt, one question at a time, no backtracking**

Assignments

- HWs due on fixed dates and times
 - **no late submission is permitted**
- All assignments should be submitted electronically on the **CodeGrade** using the link provided on the Content page in Brightspace.

Lab exercises

- Simple Coding Exercises done.
 - You have only the lab period to edit, compile and execute your solution
 - Attendance is mandatory, if you want credit
 - Demonstrate your work to Lab-TA before you leave
 - 0 – 100 points:
 - 0 - Student did not attend the lab.
 - 50 - Student attended the lab, program compiles but has major problems.
 - 75 - Student attended the lab, and program partially works (with some minor errors)
 - 100 - Student attended the lab, and program is correct

Regrading of Homework/Exams

- Please meet with a TA and arrange for re-grading.
- You have **one week** from the day grades are posted or mailed or announced
 - Late requests will not be entertained
 - Exception: final exam = we will try to grade it in one day

Academic Integrity

- You can discuss general assignment concepts with other students
- You **MAY NOT share** assignments, source code or other answers
 - Assignments are subject to manual and automated similarity checking
- If you cheat, you MAY be brought up on academic dishonesty charges without warning - we follow the university policy:
 - https://www.stonybrook.edu/commcms/academic_integrity/

Academic Dishonesty (AI-tools)

- While AI-based resources can be helpful for learning, **their use in completing programming assignments is strictly prohibited.**

Utilizing AI tools to generate or complete your code undermines the educational goals of this course and violates the university's academic integrity policy. All assignments must be your own work. Plagiarism or unauthorized assistance, including AI-generated code, will result in disciplinary action. Please remember that seeking help from teaching assistants, tutors, **or study groups is encouraged** and can be a valuable part of the learning process. However, all work submitted must be done independently to ensure you fully develop the skills needed for advanced courses and your future career. The skills you build now are crucial to your success, and adhering to these guidelines will help you achieve your academic and professional goals.

Electronic Communications

- Email sent via Brightspace is one of the ways the faculty officially communicates with you for this course. It is your responsibility to make sure that you read your email in your official University email account. If you choose to forward your official University email to another off-campus account, faculty are not responsible for any undeliverable messages to your alternative personal accounts.

Disability

- If you have a physical, psychological, medical or learning disability, contact the Student Accessibility Support Center(SASC) office at SB Union suite 107. Phone 632-6748/TDD
- **All documentation of disability is confidential**

What do you need to get started?

- Brightspace account
 - <http://mycourses.stonybrook.edu>
- Java 21:
 - <http://java.sun.com/javase/downloads>
 - API: <https://docs.oracle.com/en/java/javase/21/docs/api/>
 - The Java Tutorial: <http://java.sun.com/docs/books/tutorial>
- Eclipse IDE (v4.29.0 Released December 2023):
 - <http://www.eclipse.org/downloads>
 - Learn to use the debugger
- Liang's student Companion Website.

Tools for Writing Java Programs

- 1st Approach – the bare minimum
 - edit Java source code in text editor (ex: Notepad++ or Pico)
 - compile source code into class files from command line: `javac`
 - can be tedious
 - poor interactivity
- 2nd Approach – Integrated Development Environment (IDE)
 - combines writing, compiling, running and debugging Java code into a single application
 - Eclipse, NetBeans, IntelliJ, BlueJ, etc.
 - makes coding much more efficient and organized

How does it work?

- Java Source Code
 - you write this
 - *ProgramName.java* **files**
- Compiler Program
 - *javac ProgramName.java*
 - OR
 - **Build** - included in the Eclipse IDE
 - results Java Executable Code
 - *ProgramName.class* files - not humanly readable
- Java Virtual Machine — runs Java programs
 - *java ProgramName*
 - OR **Run** - included in the Eclipse IDE

Please

- Please be on time
- Please show respect for your classmates
- Please turn off (or use vibrate for) your cellphones

...

- On-topic questions are welcome
- Use of Zoom chat